TUTORIAL

QC/MM free energy of the SN2 step in LinB

> Draft, 2026-10-23. The figures were made with the development version of EasyHybrid 3.2.0.

Haloalkane dehalogenases break carbon–halogen bonds by an SN2 reaction. In LinB the carboxylate of **Asp108** attacks the carbon bound to the chlorine, chloride leaves, and the substrate ends up bound to the enzyme as an alkyl-enzyme (ester) intermediate:

$$\mathrm{Asp108{-}COO^- + Cl{-}CH_2{-}R \rightarrow Asp108{-}COO{-}CH_2{-}R + Cl^-}$$

In this tutorial you will:

1. scan the reaction with a QC/MM potential (AM1/AMBER);
2. run umbrella sampling (US) from the scan;
3. get the free energy profile (PMF) with WHAM;
4. **check the windows with the Umbrella Sampling Browser**, which shows that the protocol of the first version of this tutorial (k = 4000 kJ $\mathrm{mol^{-1}}$ $\mathrm{Å^{-2}}$, 15 windows) gives windows that do not overlap;
5. fix the protocol, run it again and load the structures around the transition state.

The whole tutorial runs in about 15 minutes on a 12-core workstation.

The model follows Wang *et al.*, *Phys. Chem. Chem. Phys.* **2018**, 20, 20540.

---

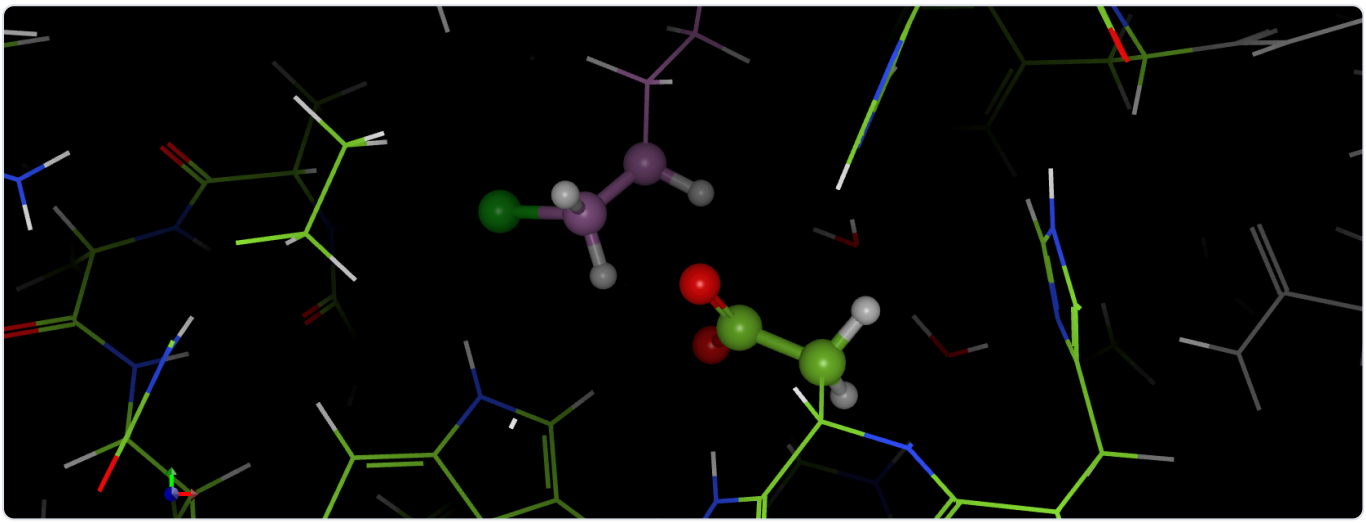
1. The system

Open `examples/tutorials/08_linb_sn2/LinB.yaml` from the EasyHybrid folder with *File* → *Open* (it can also be downloaded: LinB.yaml). The system is ready to use:

Atoms	4629 (LinB, substrate 1-chlorobutane, crystal waters)
Force field	AMBER
QC region	13 atoms, AM1, charge −1: the side chain of Asp108 (CB, CG, OD1, OD2) and the $\mathrm{Cl{-}CH_2{-}CH_2}$ end of the substrate
Link atoms	2 (Asp108 CA–CB and substrate C5–C8)
Fixed atoms	3851 (only the region around the active site moves)

The reaction coordinate uses three atoms:

Atom	Index	Name
leaving group	4609	CL1 (substrate)
central carbon	4610	C2 (substrate)
nucleophile	1670	OD1 (Asp108)



The active site: the QC region in ball-and-stick (substrate in purple, Asp108 in green)

In the reactant the C2–CL1 bond is 1.81 Å and the nucleophile is 2.78 Å from C2.

---

2. Relaxed PES scan

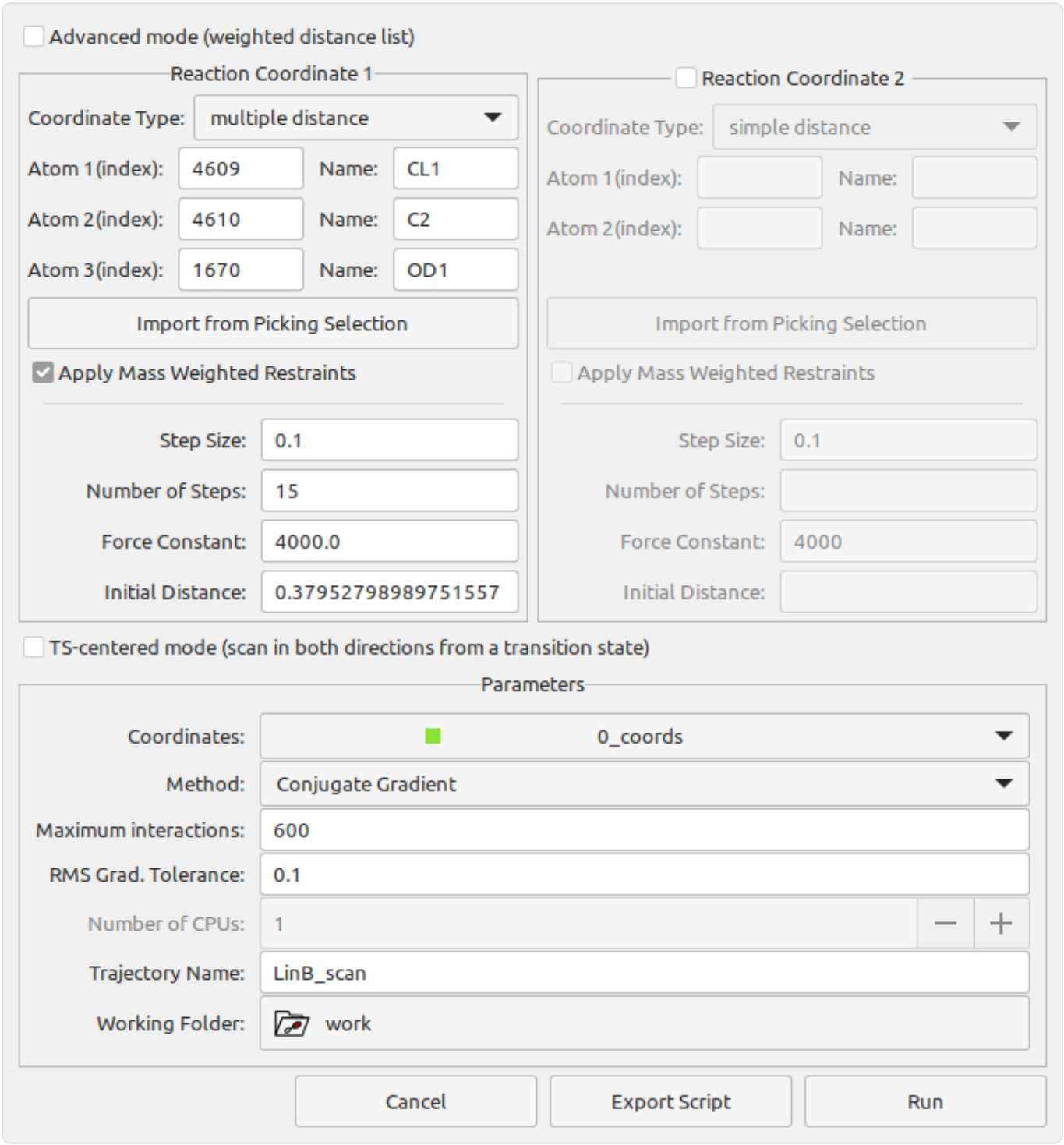
The reaction coordinate is the **multiple distance**

$$RC = \sigma_1 \cdot d(CL1\text{–}C2) - \sigma_3 \cdot d(C2\text{–}OD1)$$

that grows as the chlorine leaves and the oxygen approaches. With *Apply Mass Weighted Restraints* on, σ_1 and σ_3 are the mass weights of the two end atoms (0.689 and 0.311 here), so that the central carbon moves between them as in a real SN2 step.

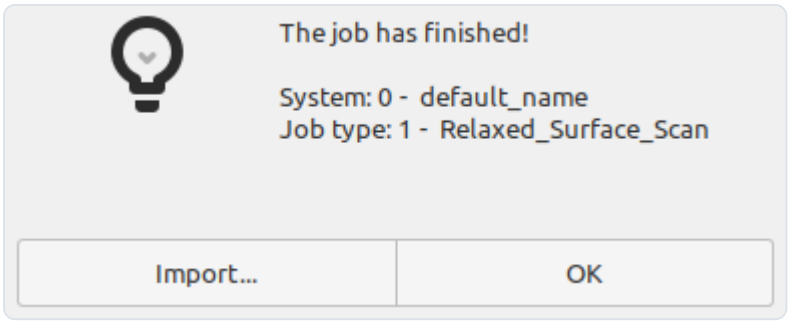
1. Pick the three atoms in the 3D view, in this order: **CL1, C2, OD1** (picking mode).
2. *Simulate* → *Reaction Coordinate Scans*.
3. Reaction Coordinate 1: Coordinate Type **multiple distance**, press *Import from Picking Selection*, and tick *Apply Mass Weighted Restraints*. The Initial Distance is filled in (0.3795 Å).
4. Step Size **0.1** Å, Number of Steps **15**, Force Constant **4000** kJ mol^{-1} $Å^{-2}$.
5. Parameters: Method **Conjugate Gradient**, Maximum interactions **600**, RMS Grad. Tolerance **0.1**. Trajectory Name `LinB_scan`.

6. *Run*.

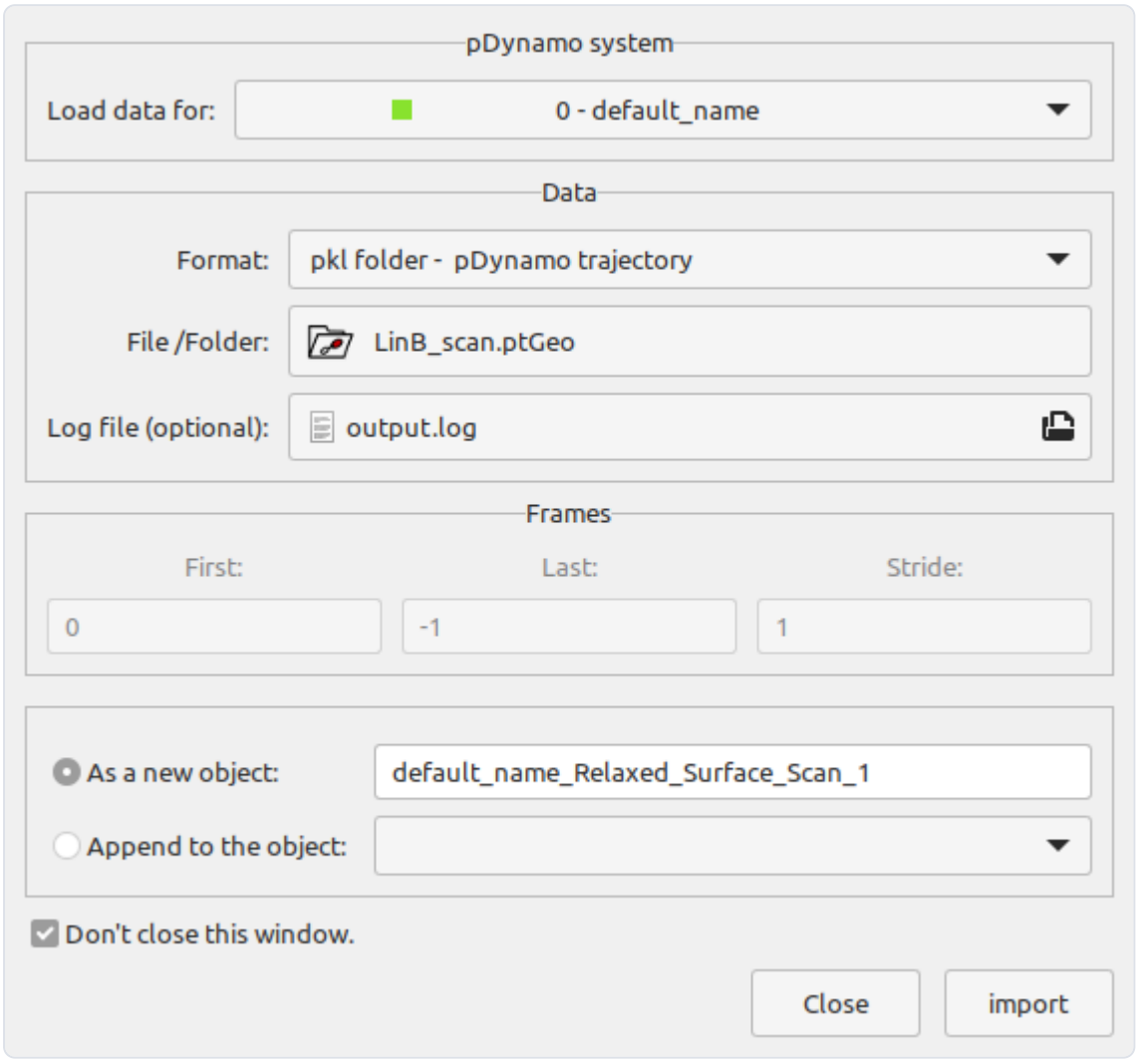


The scan window

The scan takes about 20 s. When it ends, the *Finished!* dialog offers *Import...*, which opens *Import Data* with the trajectory folder and its `output.log` filled in; press *import*.

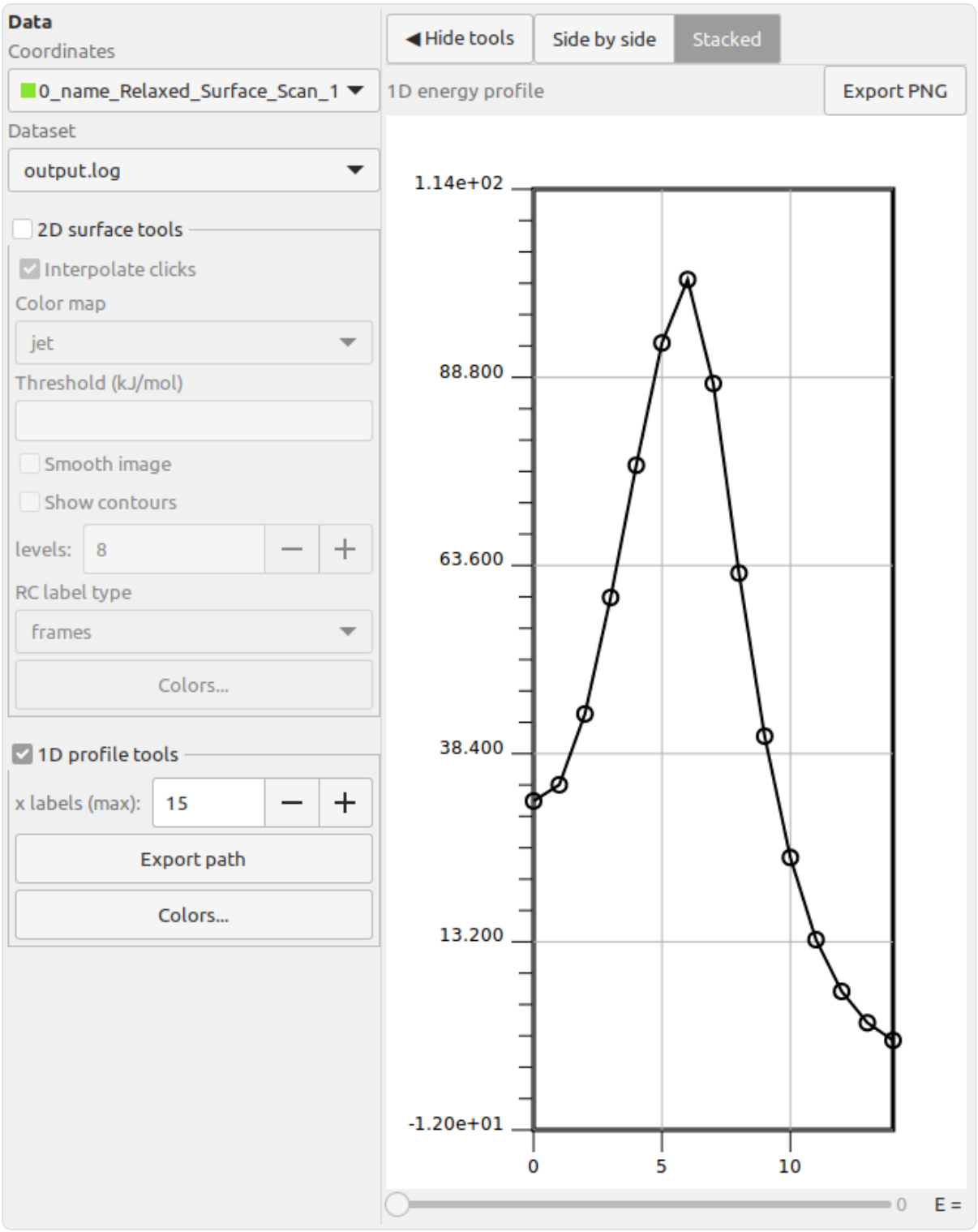


The job has finished: Import... loads the result

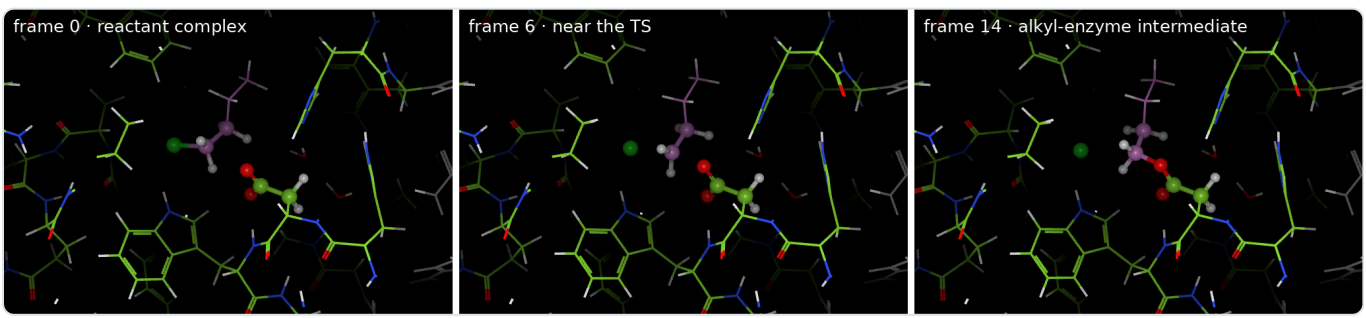


Import Data, filled in from the job

The new object has 15 frames. *Analysis* → *PES Analysis* plots the energy along the scan: the barrier is about **70 kJ mol^{-1}** (frame 6 or 7) and the intermediate lies about 32 kJ mol^{-1} below the reactant.



PES analysis of the scan

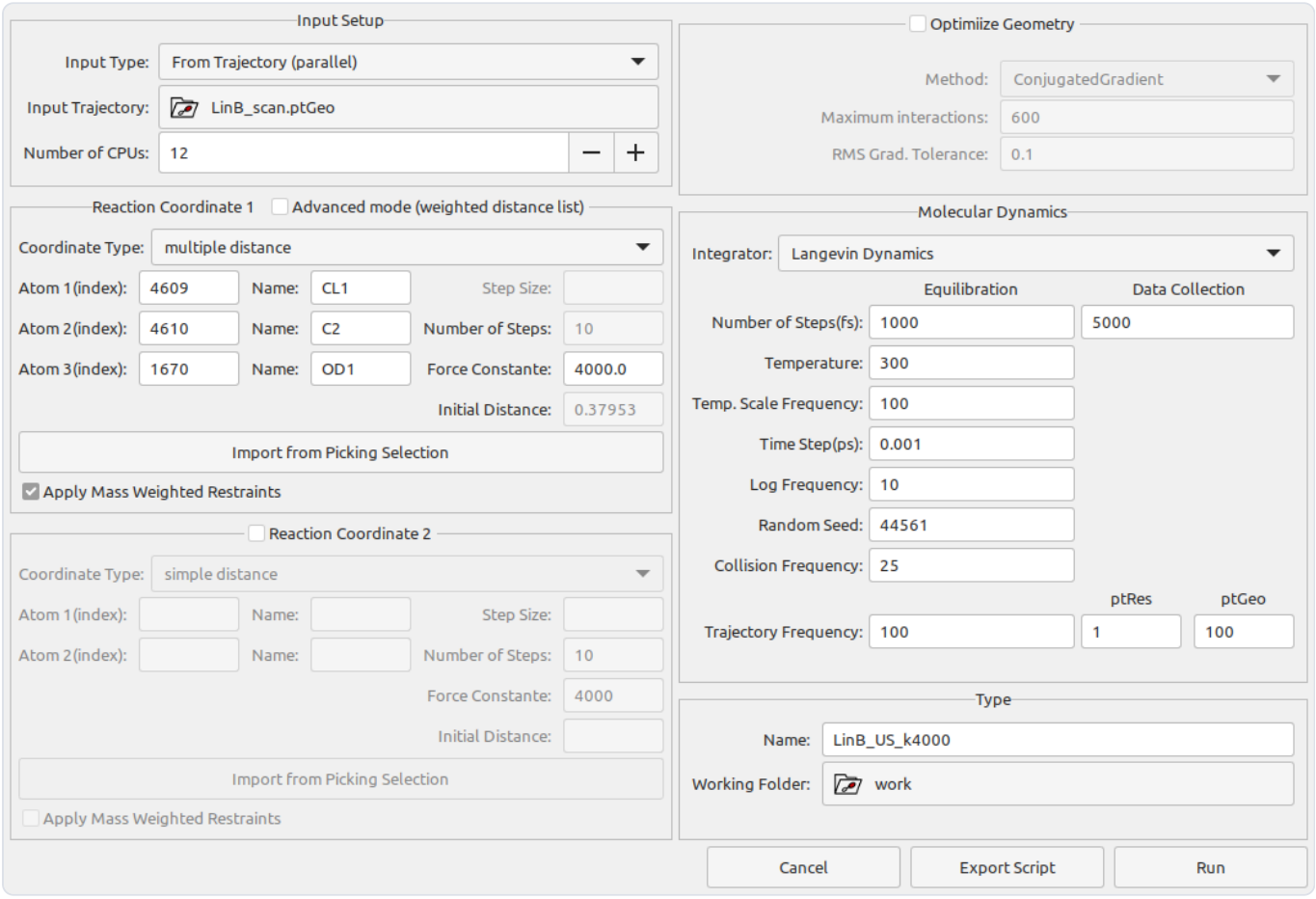


Three frames of the scan: the chloride leaves as OD1 bonds to C2

3. Umbrella sampling

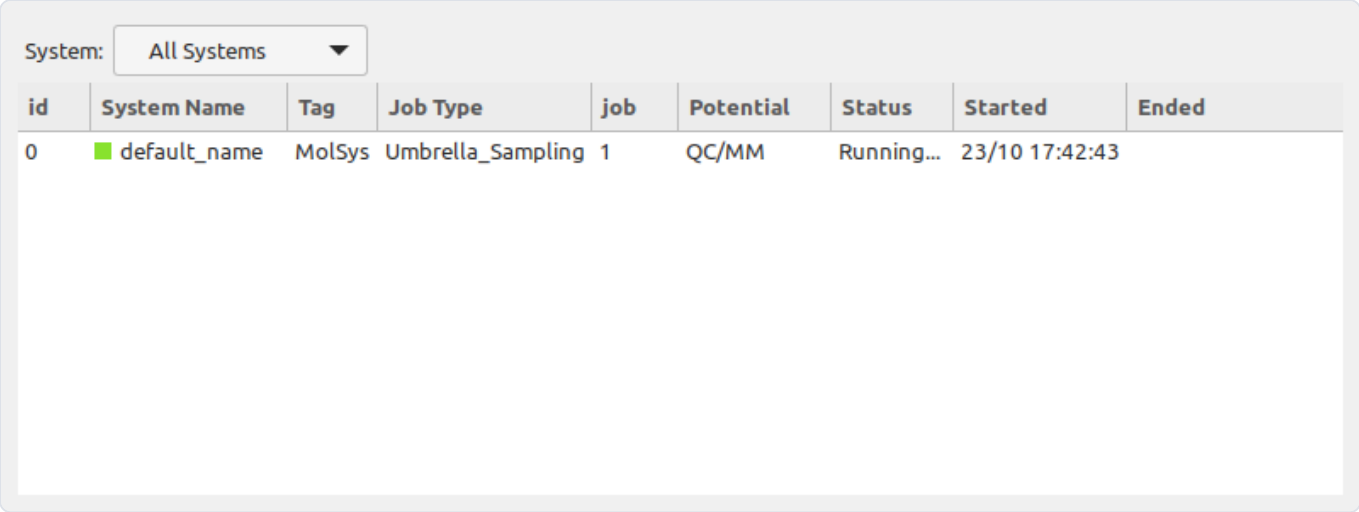
Each US window starts from one frame of the scan and is restrained to the RC value of that frame.

1. *Simulate* → *Umbrella Sampling*.
2. Input Type **From Trajectory (parallel)**; Input Trajectory: the `LinB_scan.ptGeo` folder. EasyHybrid reads the reaction coordinate from the scan's `output.log` (atoms, type, mass weights and initial value), so Reaction Coordinate 1 is already filled in.
3. Number of CPUs: as many as you have (12 here; the 15 windows run in parallel).
4. Force Constant **4000** kJ mol^{-1} Å^{-2} (the value of the first version of this tutorial).
5. Molecular Dynamics: Langevin, 300 K, time step 0.001 ps, collision frequency 25 ps^{-1}. Equilibration **1000** steps, Data Collection **5000** steps; ptRes every step, ptGeo every 100.
6. Name `LinB_US_k4000`, *Run*.



The umbrella sampling window, from the scan trajectory

The run takes about 3 min with 12 processes. The *Process Manager* shows it while it runs.



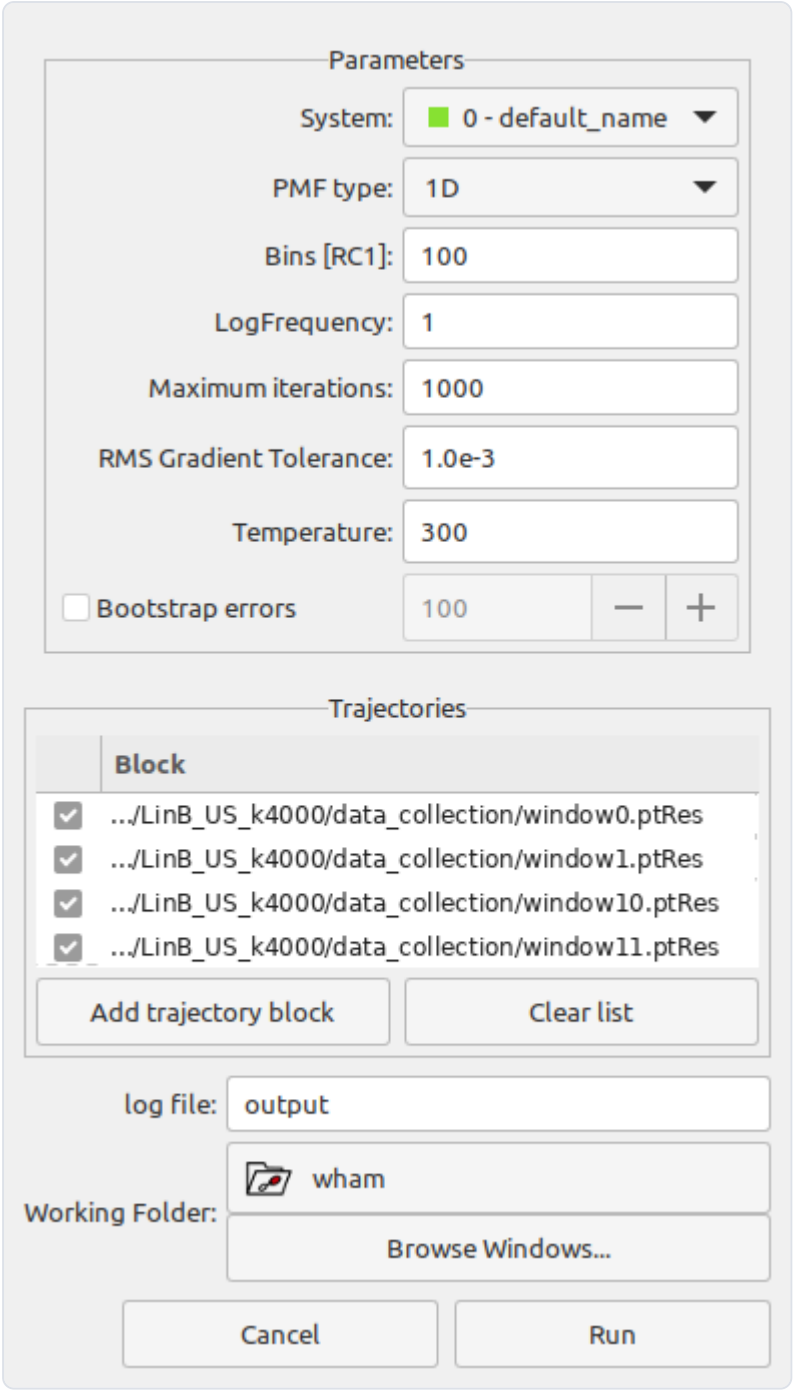
The Process Manager during the run

The run folder has one sub-folder per stage (`equilibration` , `data_collection` , `data_collection_traj`) with one file per window.

---

4. WHAM

Analysis → *WHAM*: add the `.ptRes` files of `LinB_US_k4000/data_collection` , Bins **100**, Temperature **300**, Working Folder `LinB_US_k4000/wham` , *Run*. WHAM writes `output_pmf.log` .



The WHAM window

The PMF has a barrier of about 85 kJ mol^{-1}. Before trusting it, check the windows.

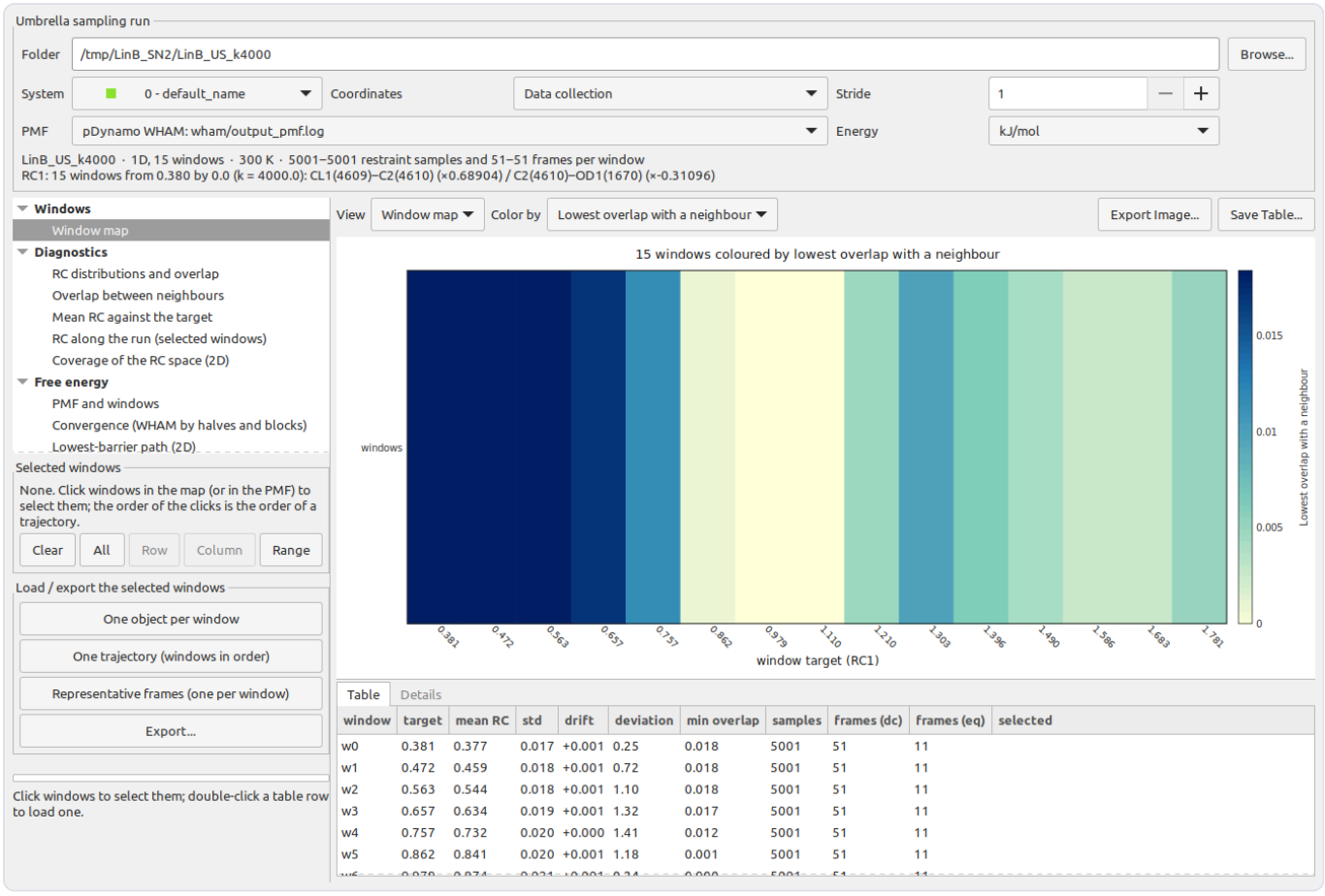
---

5. Checking the windows: the Umbrella Sampling Browser

Analysis → Umbrella Sampling Browser... (or *Browse Windows...* in the WHAM window) and choose the run folder `LinB_US_k4000`. The browser reads the windows, their targets and force constants, and finds the PMF in `wham/`. The US browser tutorial describes all of its views; here we use the ones that matter for this run.

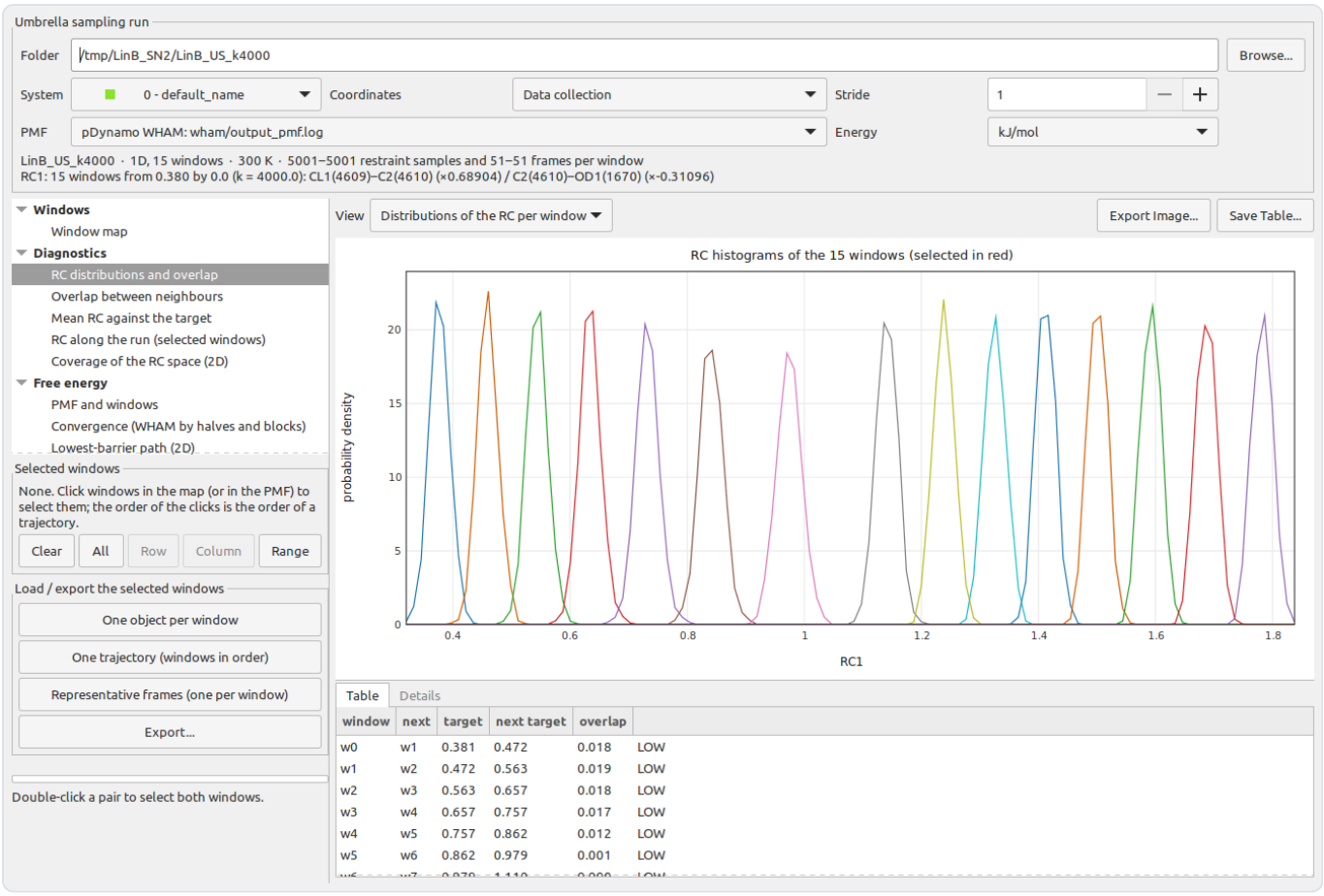
The windows do not overlap

Window map, coloured by *Lowest overlap with a neighbour*: the whole strip is at the bottom of the scale.



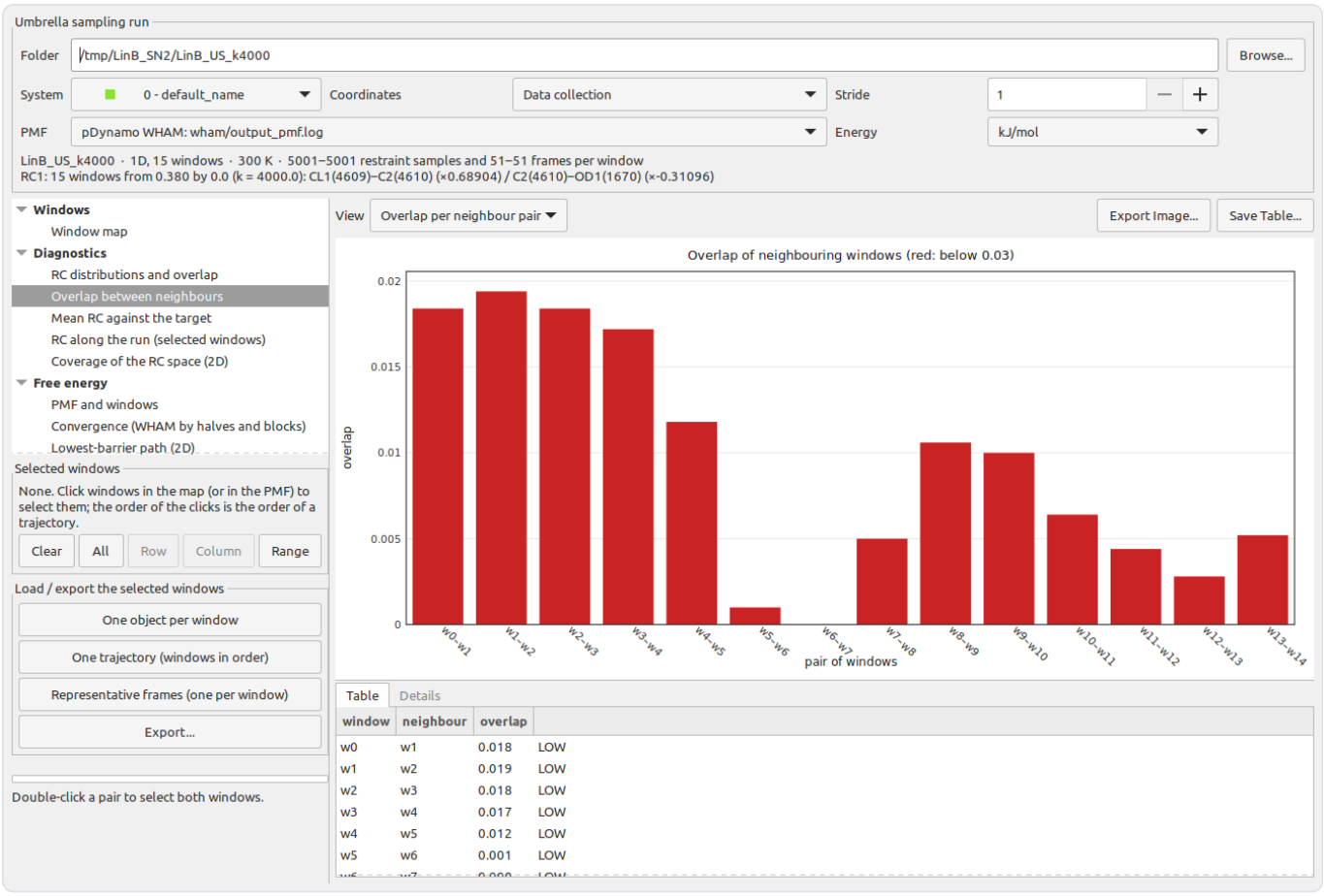
Window map of the k = 4000 run, coloured by the lowest overlap with a neighbour

RC distributions and overlap: each window is a narrow peak with gaps between the peaks.



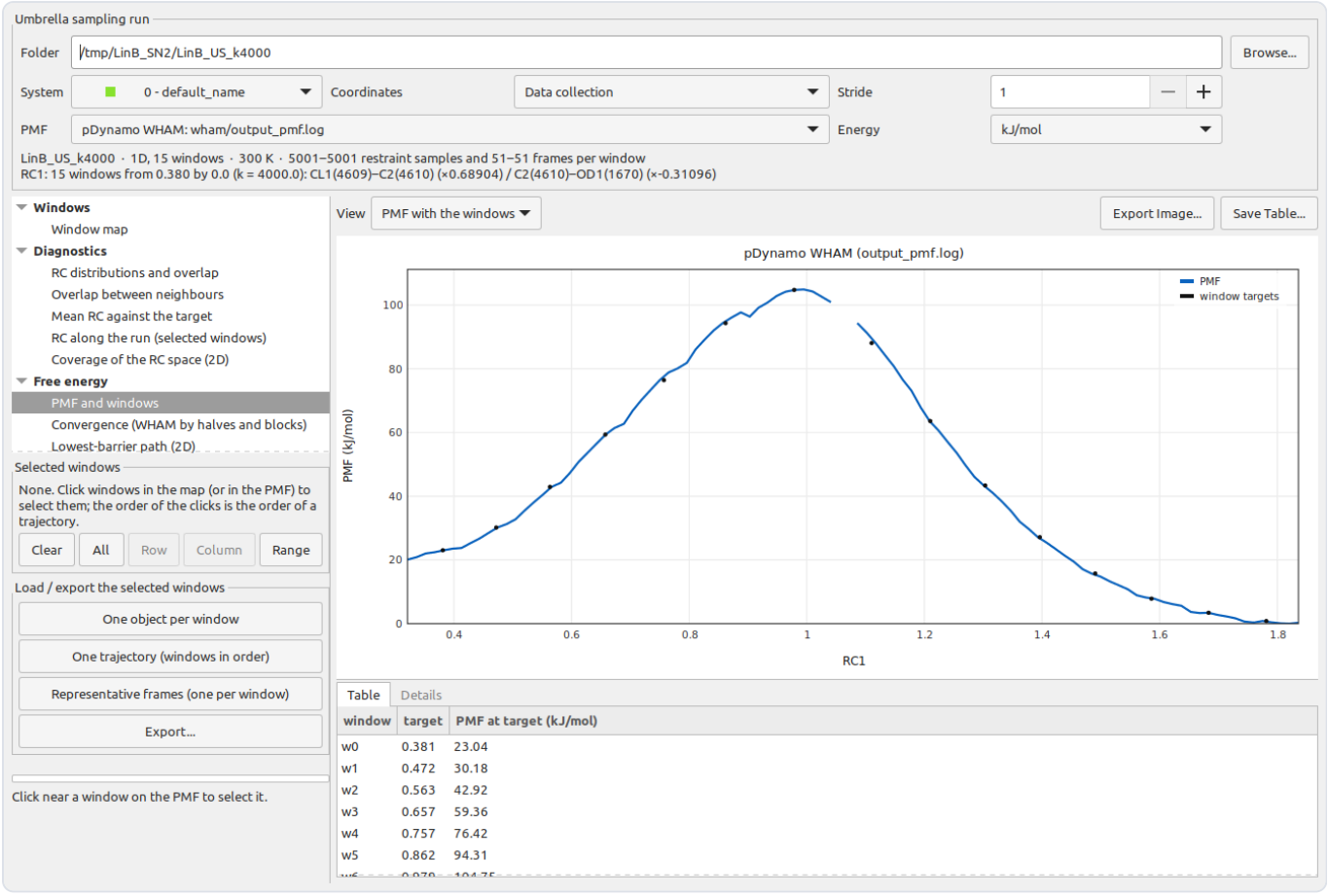
RC distributions of the k = 4000 run

Overlap between neighbours: **all 14 pairs are below 0.03**, the largest is 0.019, and windows 6 and 7 (at the transition state) share no samples at all.



Overlap per pair of neighbours, k = 4000: every bar is red

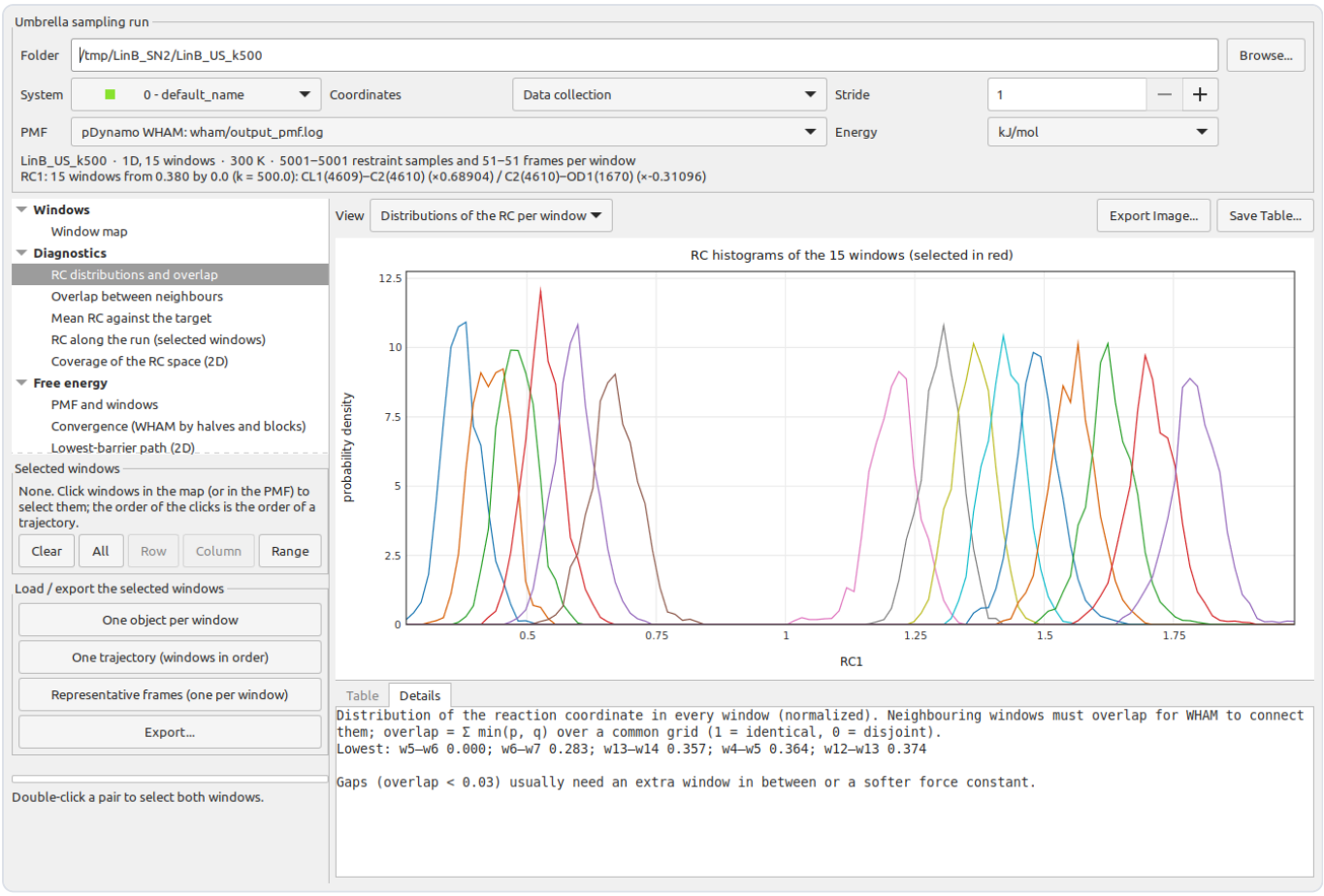
The reason is simple. A harmonic restraint of force constant k lets the RC fluctuate by about √(kT / 2k): **0.018 Å for k = 4000** at 300 K, while the windows are 0.1 Å apart. WHAM still returns a smooth curve, but with no overlap the data do not fix the relative free energy of neighbouring windows, so the height of that curve is not reliable.



The PMF of the k = 4000 run: a smooth curve from disconnected windows

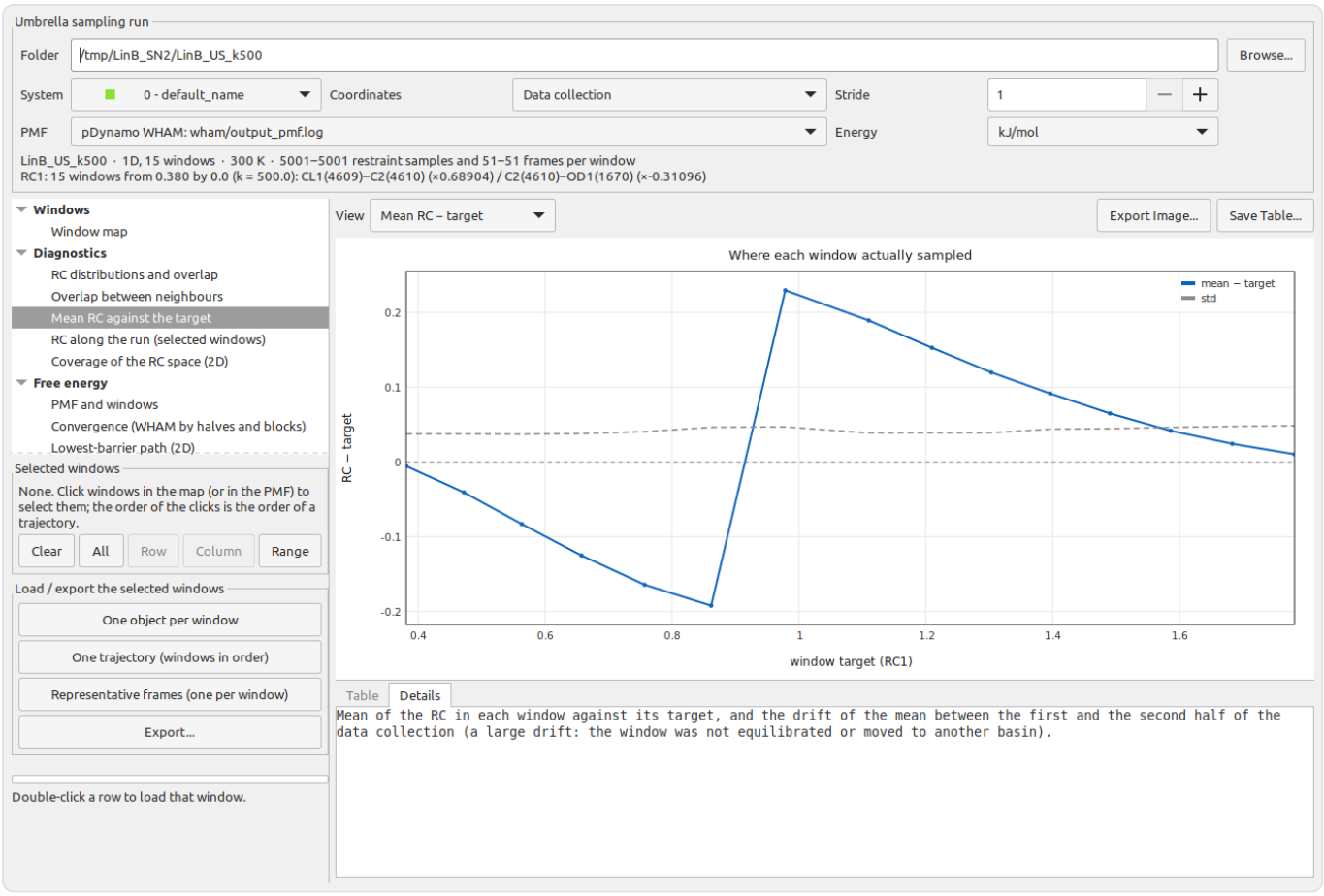
A softer restraint is not enough either

With k = 500 the RC fluctuates by 0.05 Å, enough for 0.1 Å spacing. The run `LinB_US_k500` (same settings, only the force constant changed) shows the opposite problem:



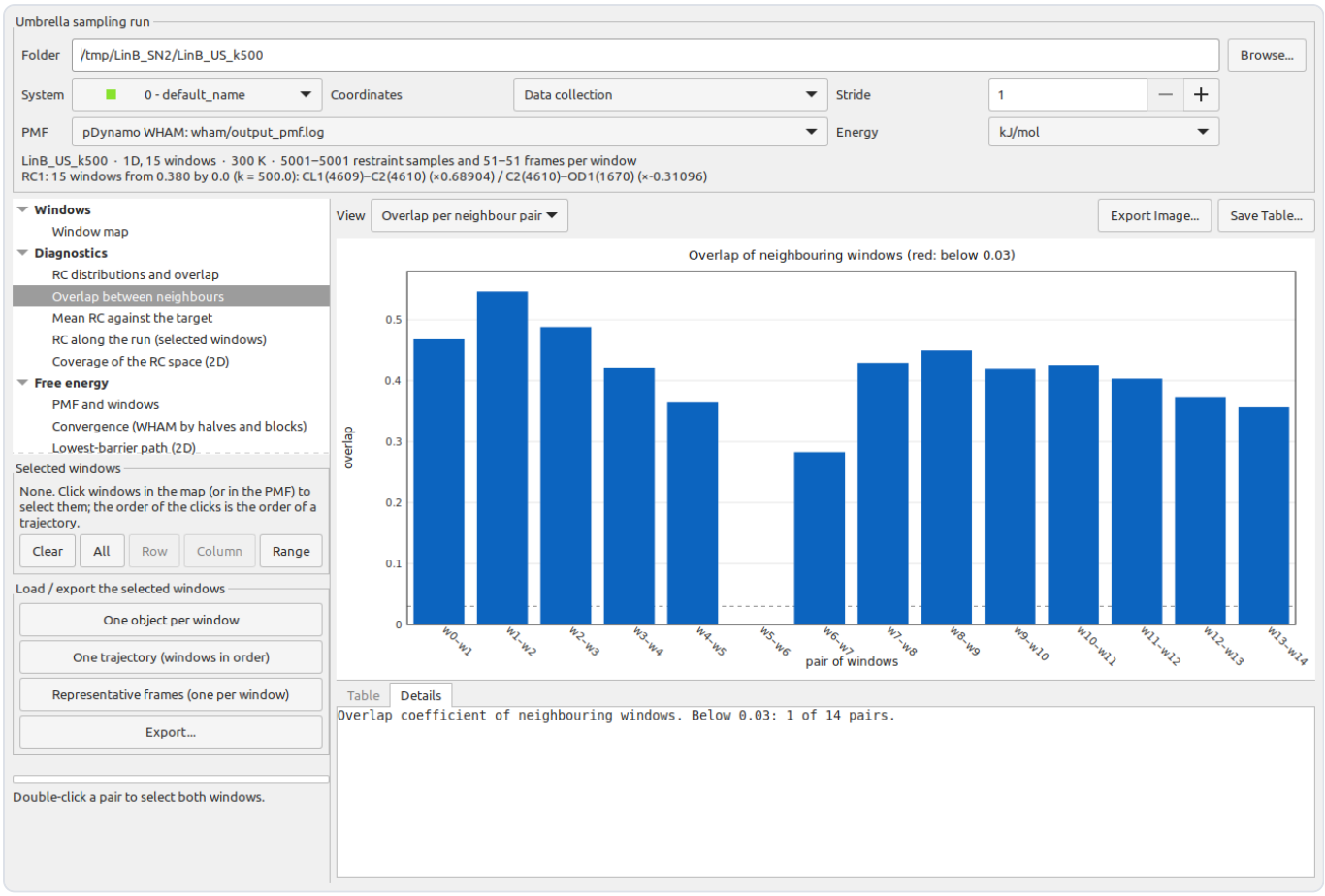
RC distributions of the k = 500 run: a gap in the middle

Mean RC against the target shows why: the windows before the barrier slide back towards the reactant and those after it slide forward towards the product, by up to 0.23 Å. The restraint is too weak for the slope of the PMF near the transition state.

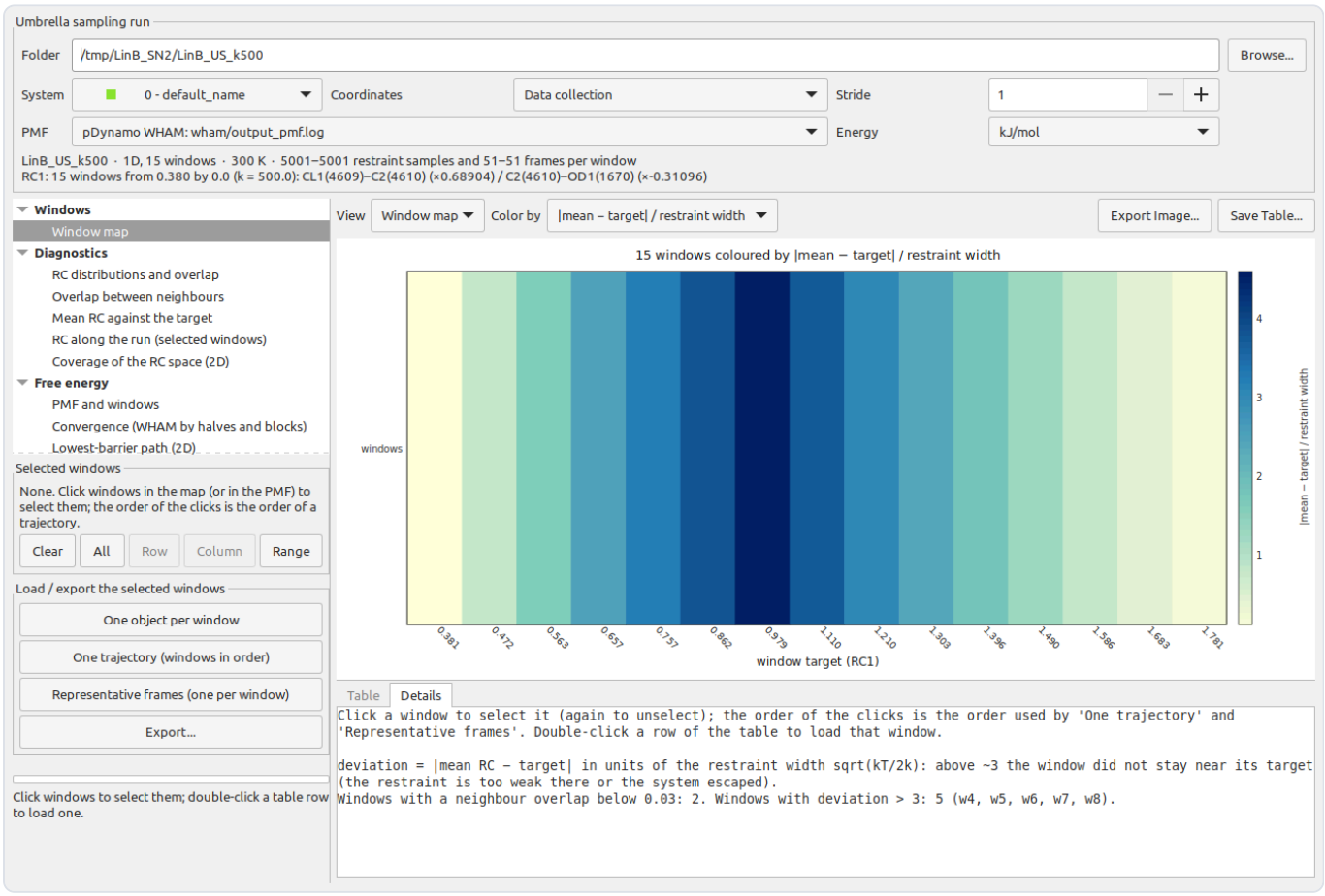


Mean RC − target for the k = 500 run

The browser counts 5 windows (w4–w8) that drifted more than 3 restraint widths, and the pair w5–w6 does not overlap: **the transition state was never sampled**. WHAM gives a barrier of only 62 kJ mol^{-1} here, which is wrong.



Overlap per pair, k = 500: one pair at zero, right at the barrier



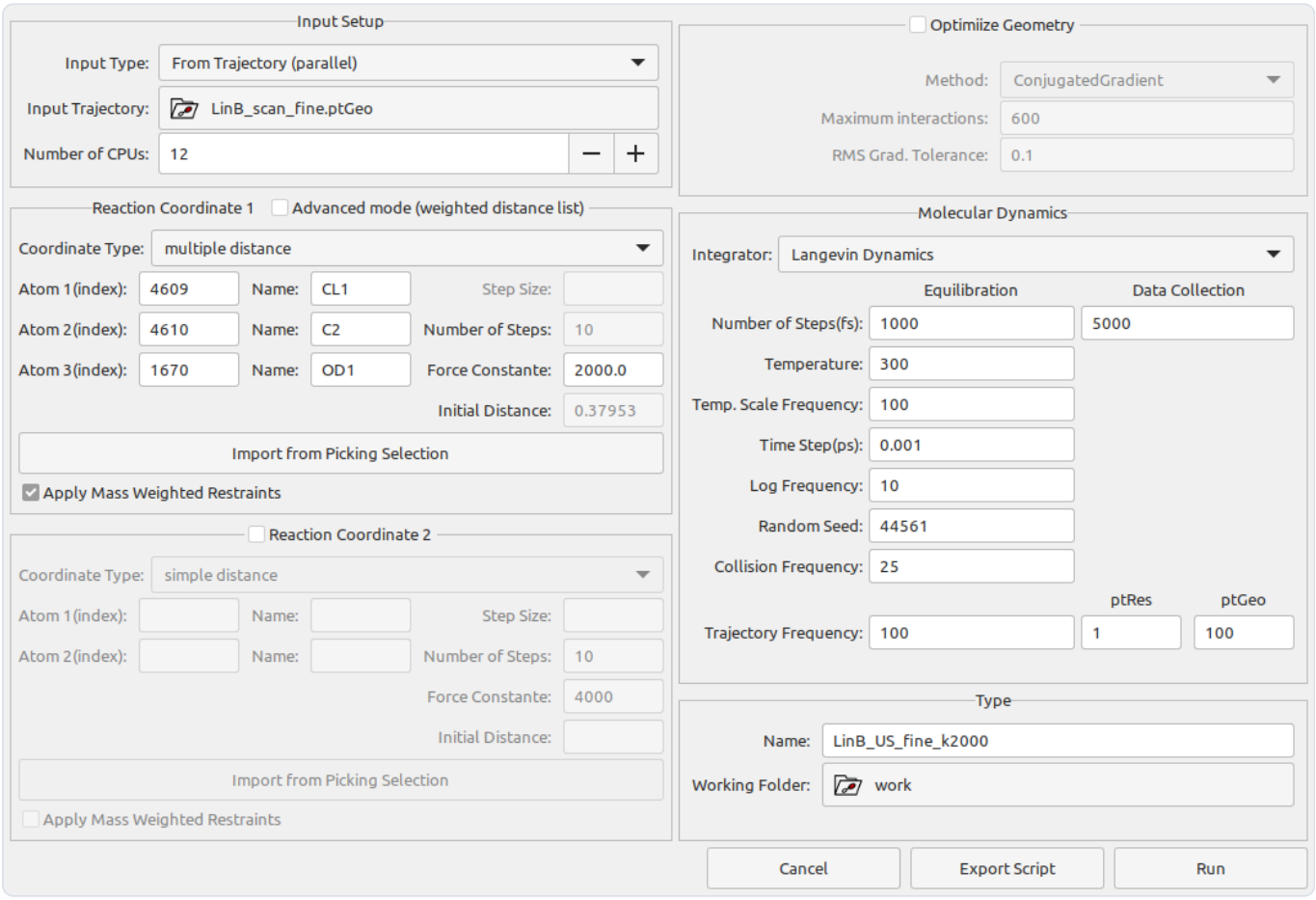
Window map of the k = 500 run, coloured by the deviation from the target

6. A protocol that works

The fix is to place windows closer together **and** keep the restraint strong enough to hold them on the barrier:

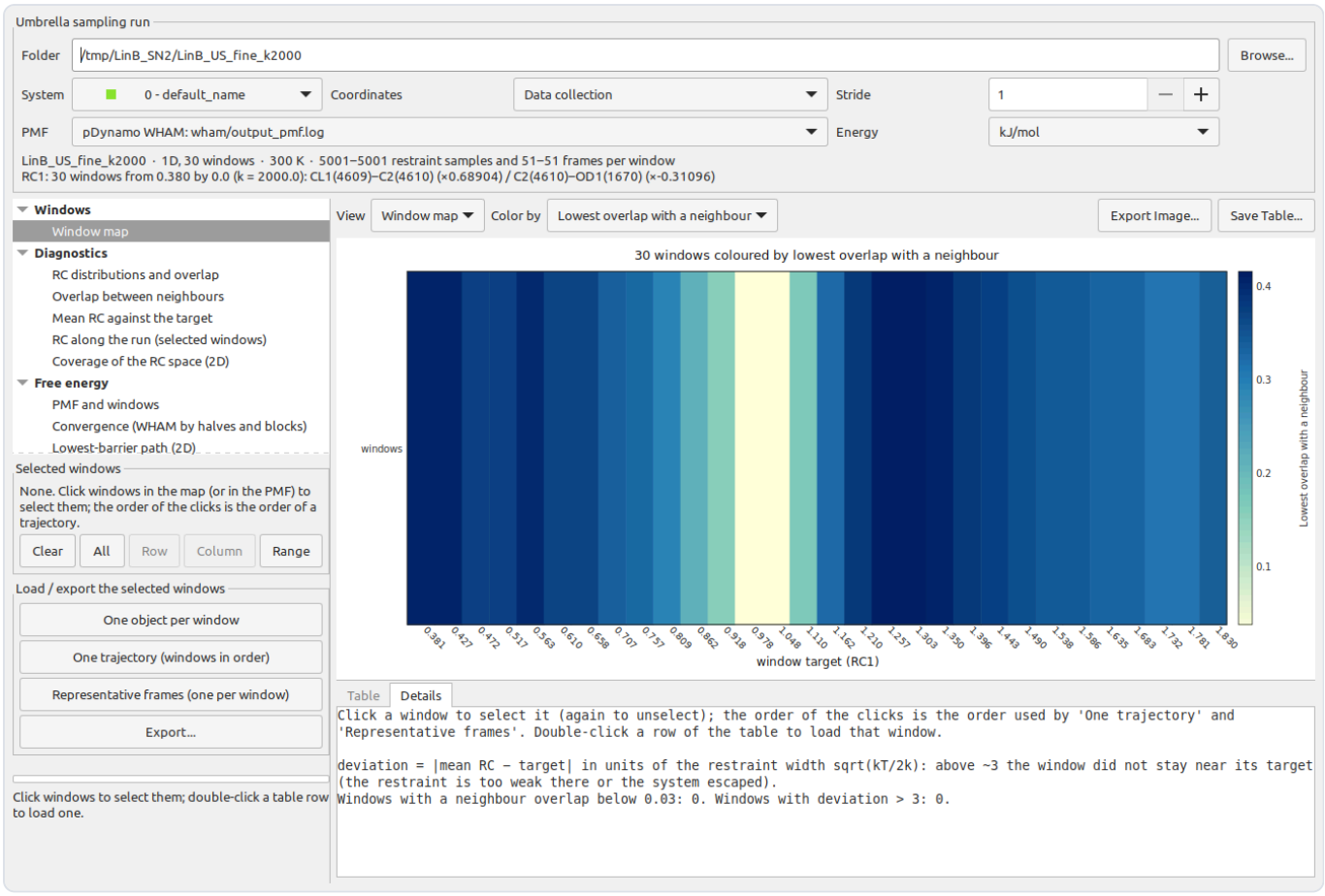
	First version	Fixed
Scan	15 steps of 0.1 Å	**30 steps of 0.05 Å**
US force constant	4000	**2000** kJ mol^{-1} Å^{-2}
Restraint width √(kT / 2k)	0.018 Å	0.025 Å
Windows	15	30

Repeat section 2 with Step Size 0.05 and Number of Steps 30 (Trajectory Name `LinB_scan_fine`), then section 3 from that scan with Force Constant 2000 (Name `LinB_US_fine_k2000`), then WHAM. The 30 windows take about 4 min with 12 processes.

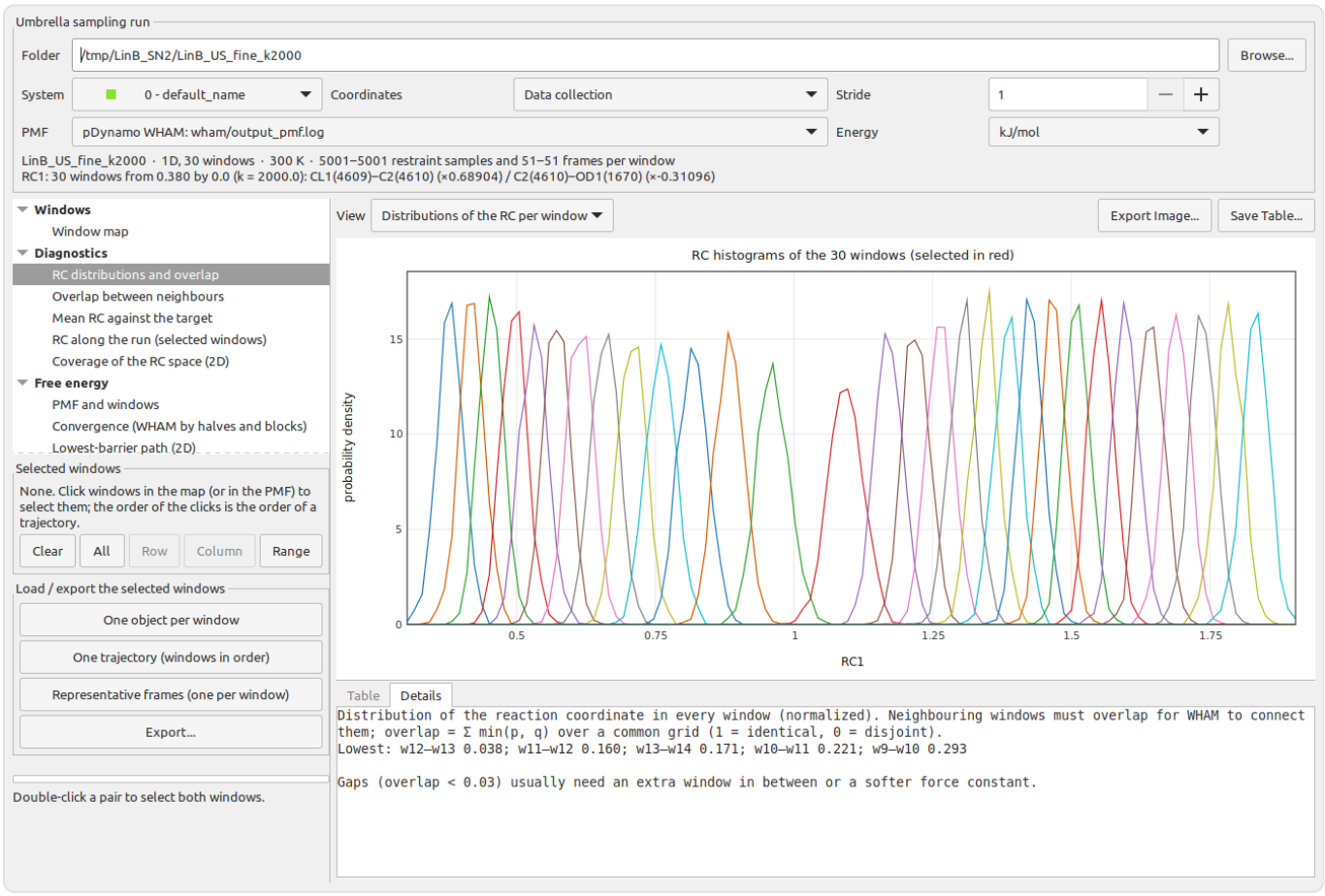


Umbrella sampling from the fine scan, k = 2000

Open the new run in the browser.

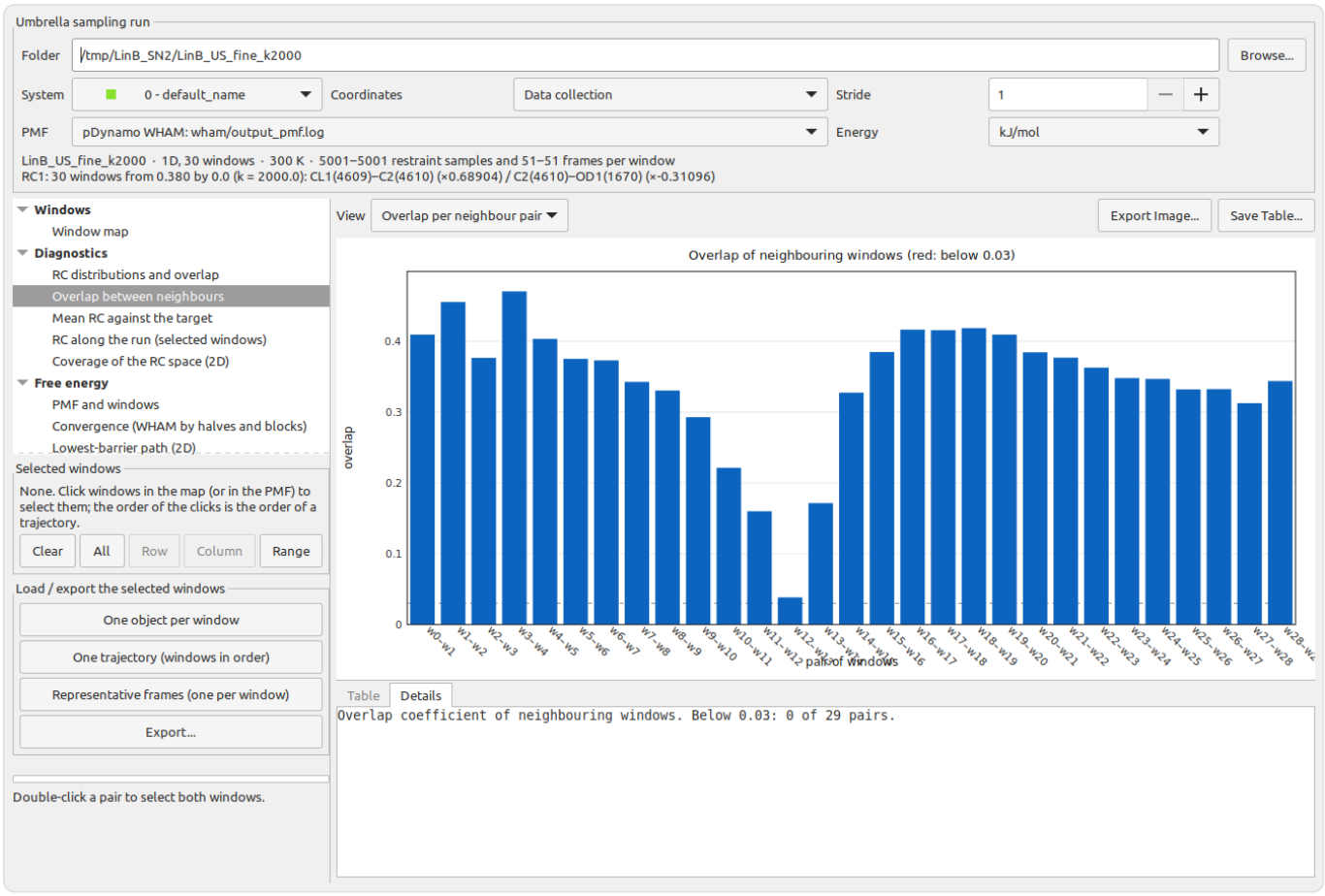


Window map of the fixed run, coloured by the lowest overlap with a neighbour



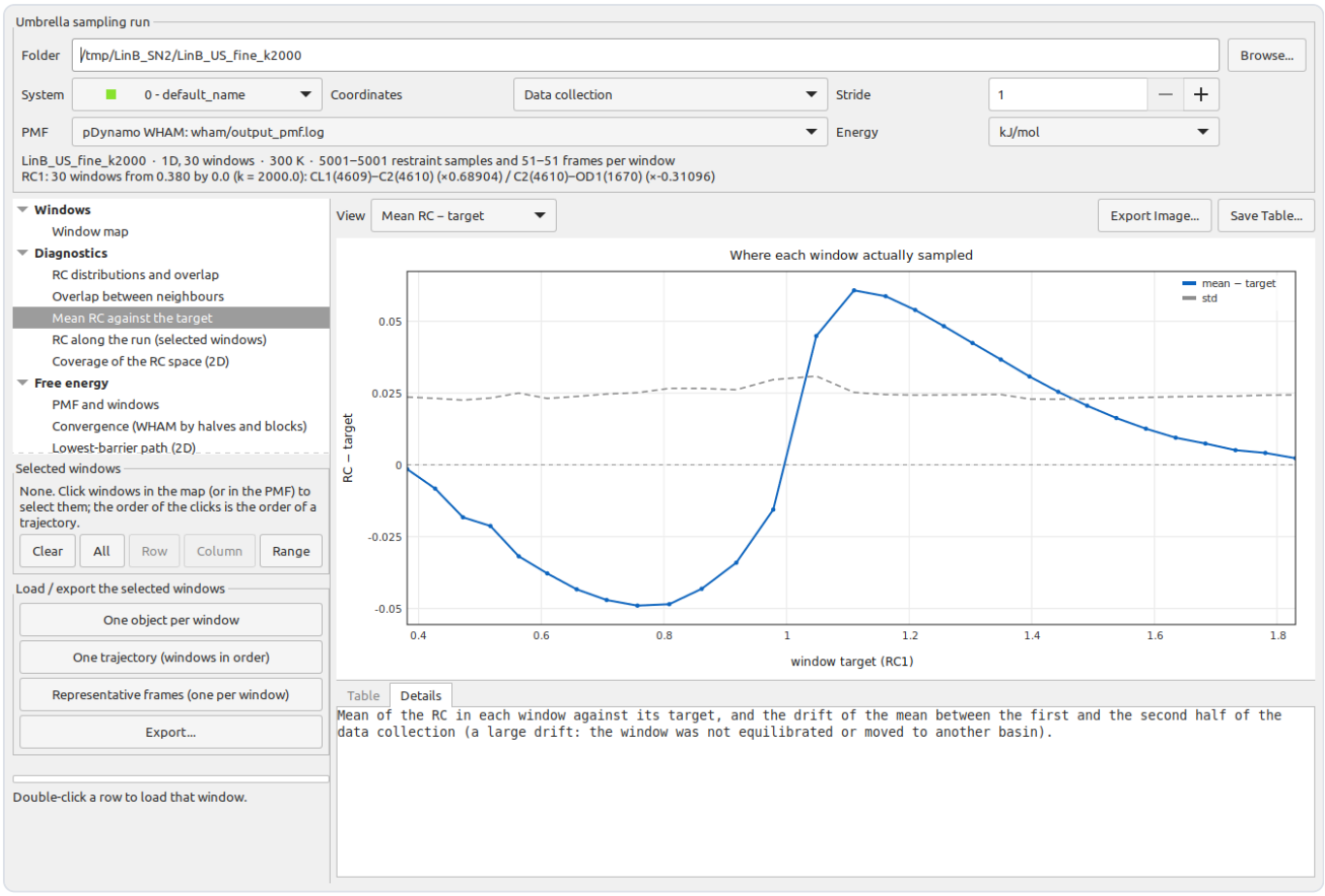
RC distributions of the fixed run: every window overlaps its neighbours

Now **0 of 29 pairs are below 0.03**. Most pairs overlap by 0.3–0.5; the weakest one is w12–w13 at the transition state (0.04), where the PMF is steepest. It is just enough; for a production run add a window between them or run those two windows longer.

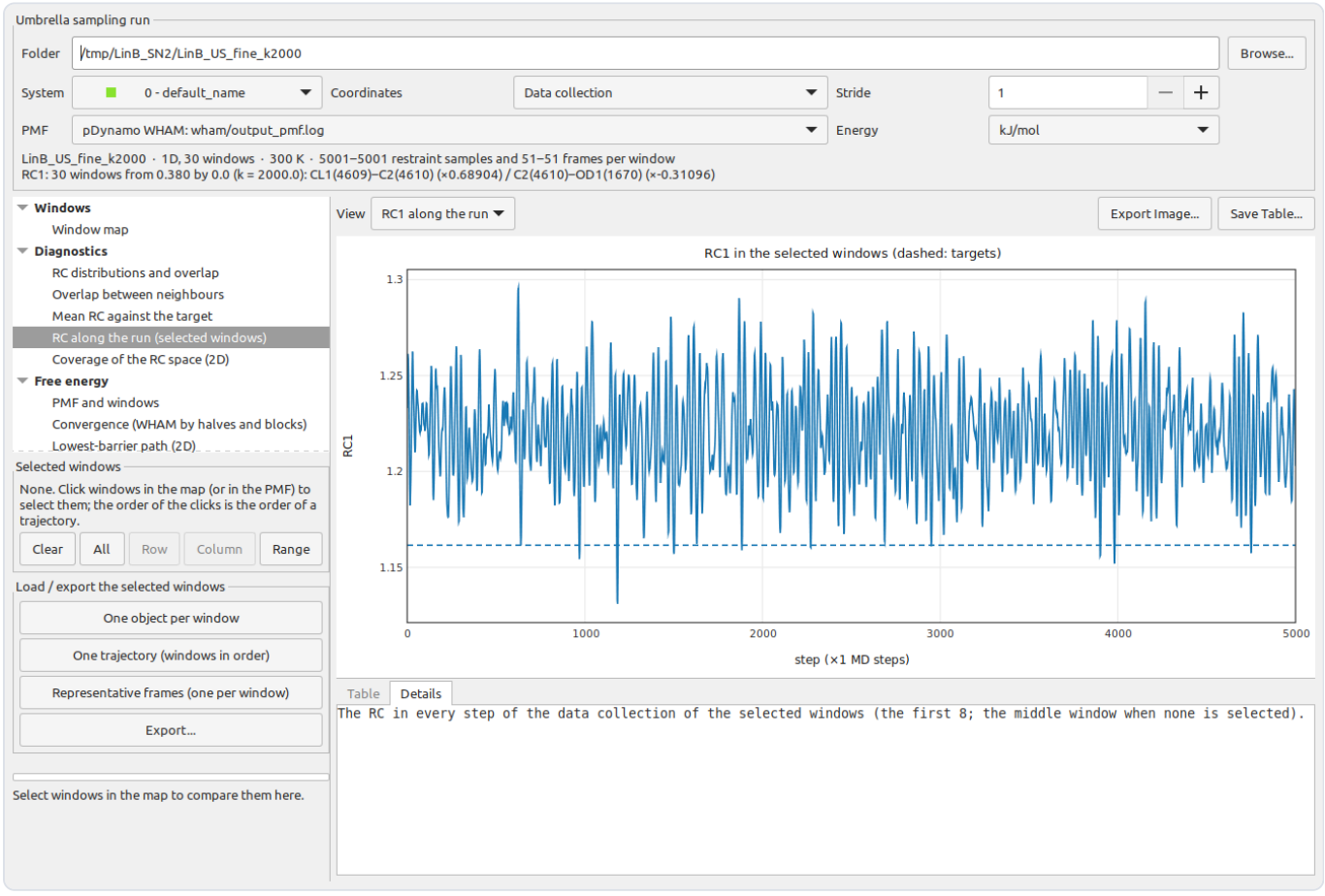


Overlap per pair, fixed run

No window drifted by more than 2.5 restraint widths:

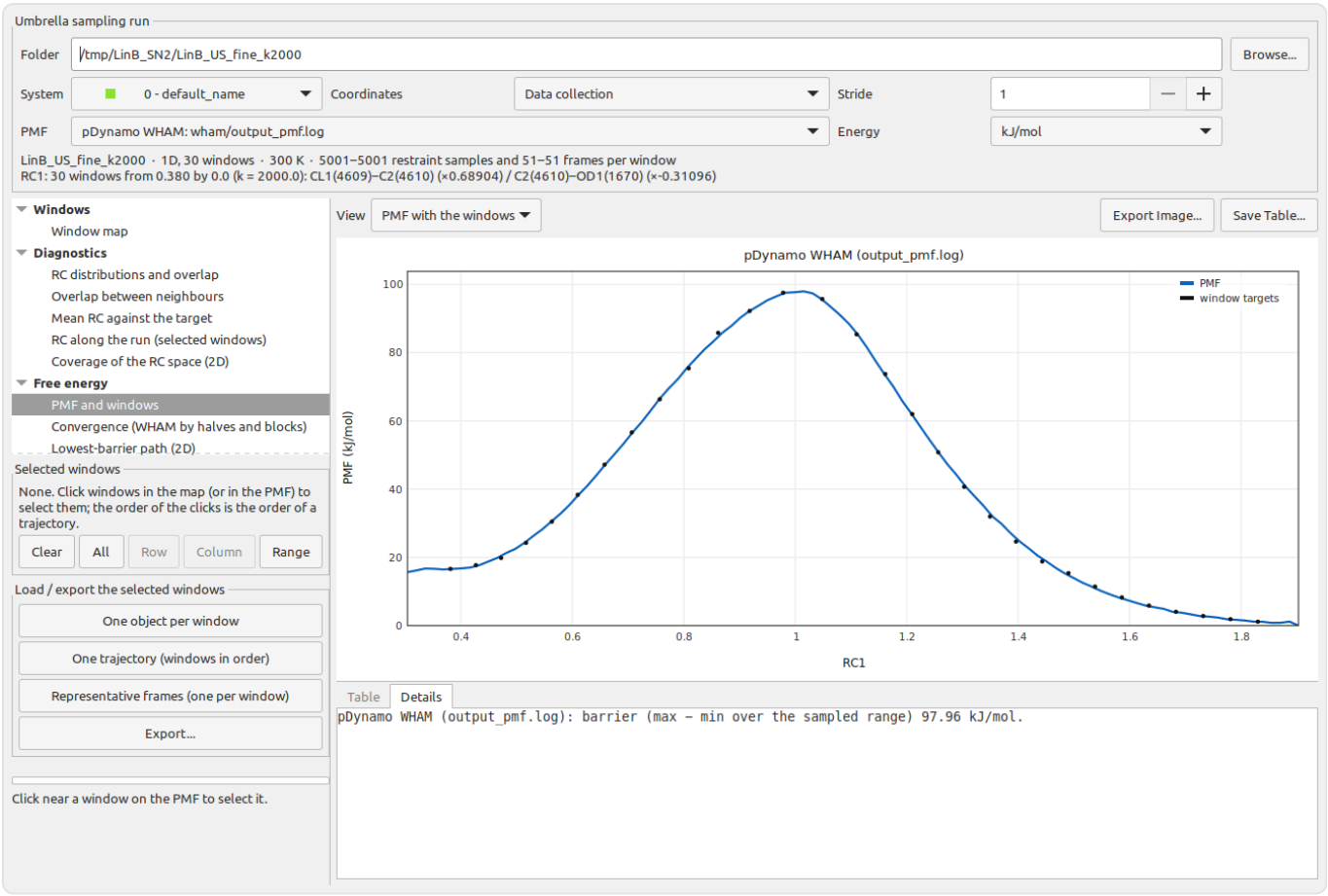


Mean RC − target, fixed run



The RC along the data collection

The free energy



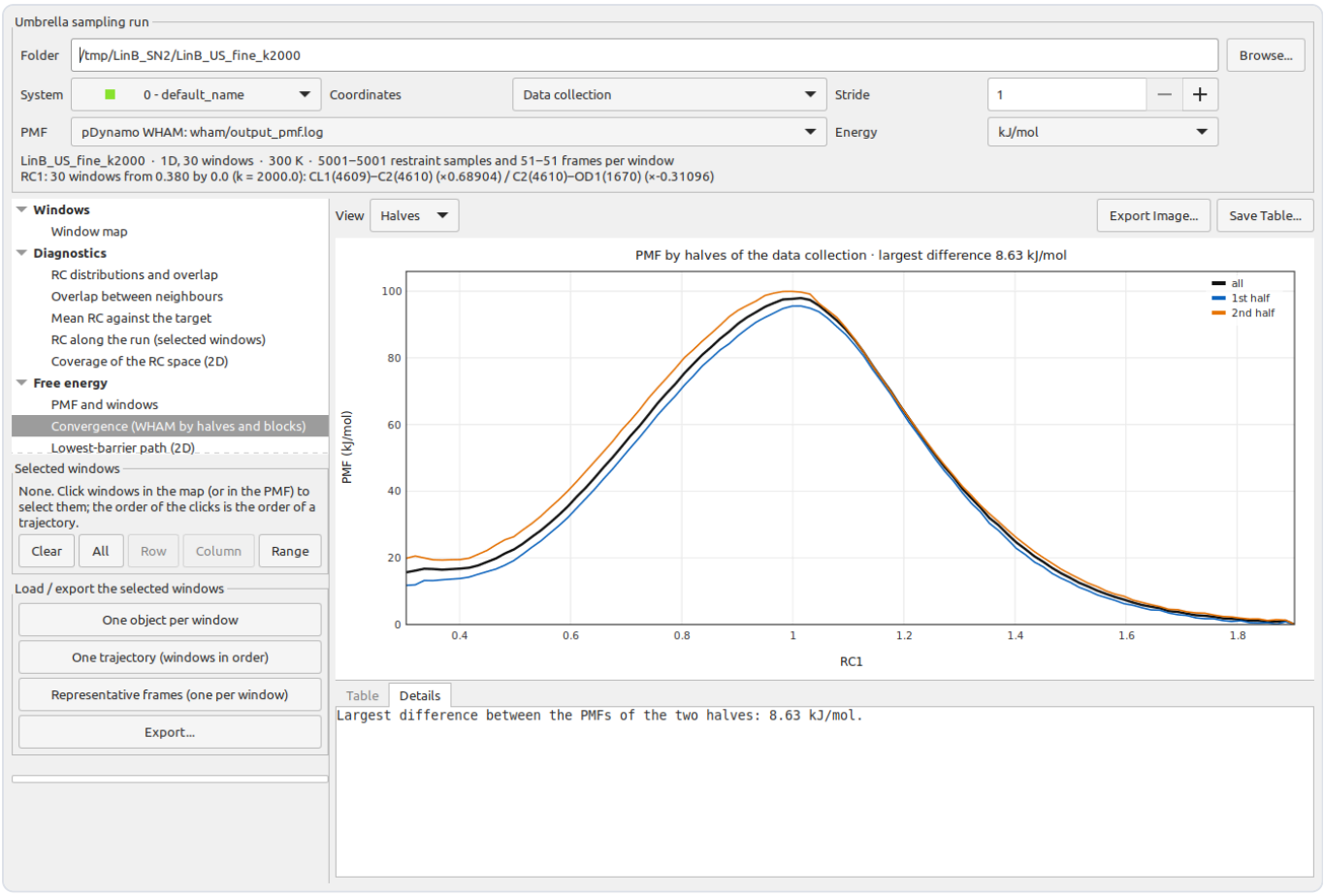
PMF of the fixed run with the window targets

The barrier is **82 kJ mol^{-1}** (19.7 kcal mol^{-1}) from the reactant minimum (RC ≈ 0.30 Å) to the transition state (RC ≈ 1.01 Å), and the intermediate lies about 16 kJ mol^{-1} below the reactant. These are AM1/AMBER numbers from a short run; compare them with higher-level QC/MM results (see the reference above) before drawing conclusions.

Run	Windows	k	Lowest overlap	Barrier (kJ mol^{-1})	Verdict
`LinB_US_k4000`	15	4000	0.000	85	windows disconnected
`LinB_US_k500`	15	500	0.000	62	TS not sampled
`LinB_US_fine_k2000`	30	2000	0.04	**82**	usable

Is 5 ps per window enough?

Convergence (press *Compute*) runs WHAM on the first and the second half of the data collection. The halves differ by up to 8.6 kJ mol^{-1}, mostly on the reactant side (the second half is higher); the barrier itself changes less. For a publication, collect at least four times longer and check this plot again.

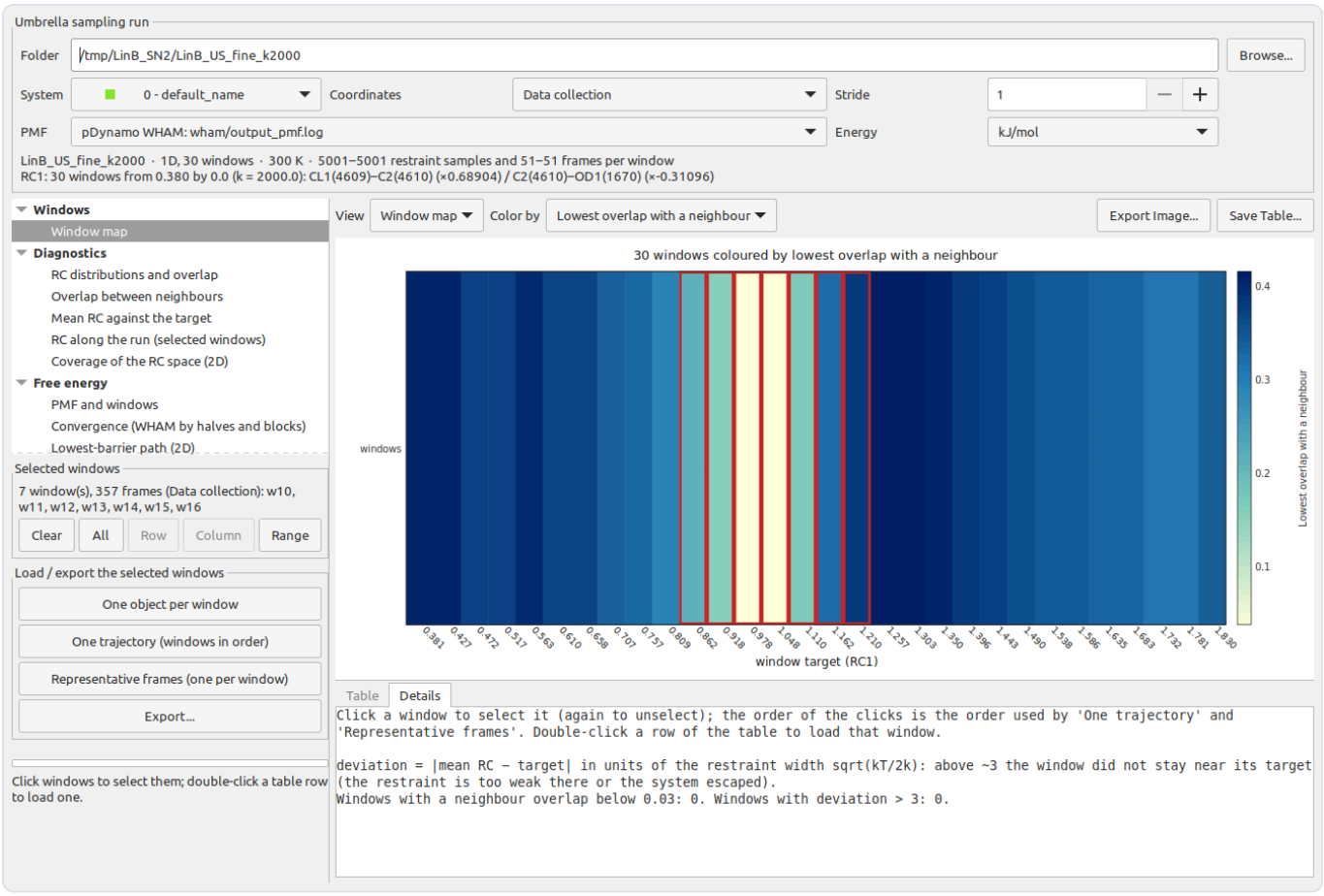


PMF of the two halves of the data collection

---

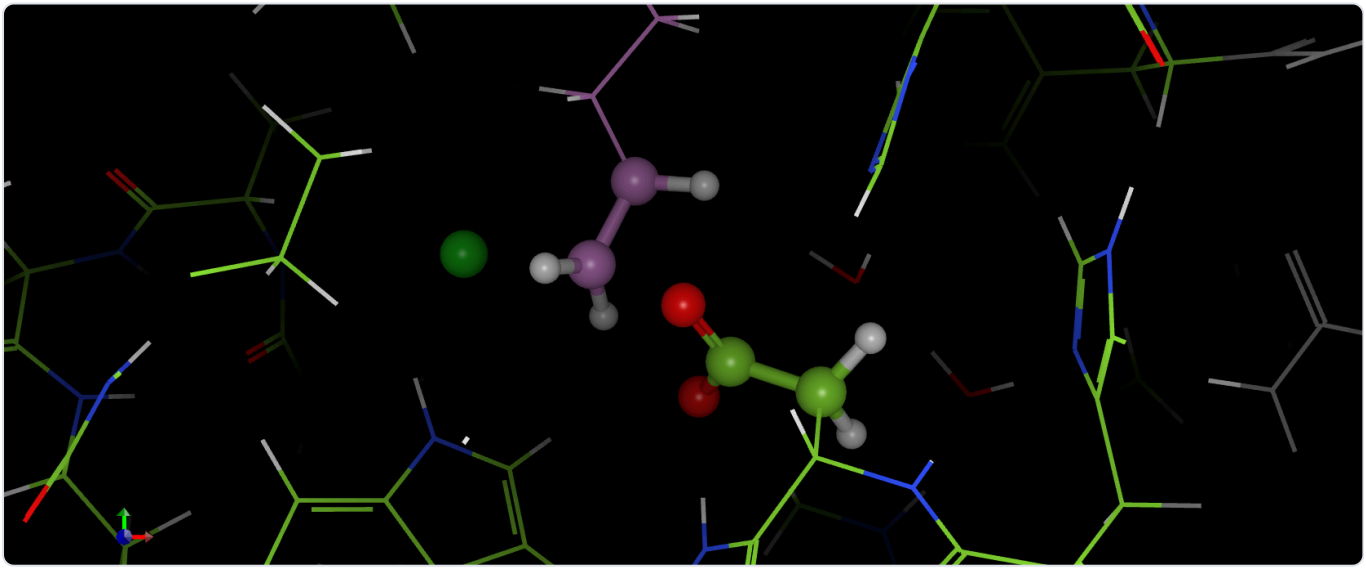
7. Structures around the transition state

In the *Window map*, click w10 and then w16 and press *Range*: the seven windows around the barrier are selected.



Windows w10–w16 selected on the map

Representative frames (one per window) loads one object with one frame per selected window (the frame whose RC is closest to the target of that window). Play it with the mini player to watch the chloride leave and OD1 bind to C2.



Representative frame of window w13: chloride leaving, OD1 forming the bond to C2

One trajectory (windows in order) loads every frame of the selected windows as one trajectory, and *Export...* writes them as DCD, XYZ or PDB for analysis elsewhere.

8. Summary

- Run a PES scan, then US from the scan frames, then WHAM.
- **Always open the run in the Umbrella Sampling Browser before trusting the PMF.**
- Neighbouring windows must overlap: the restraint width $\sqrt{(kT / 2k)}$ should be comparable to the window spacing (here k = 2000 with 0.05 Å, not k = 4000 with 0.1 Å).
- Windows that slide away from their target near the barrier mean the restraint is too weak there.
- Compare the PMF of the two halves of the data to see whether the run is long enough.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3